Project Overview

Overview of Project 1

1. Brief Description

This study explores if paying financial incentives to community-based volunteers (CBVs) has an impact on improving maternal health behaviors, specifically antenatal care (ANC) timing and location, in rural Zambia. Using a matched-pair cluster randomized trial, 98 rural health facilities were assigned to either a treatment group—where CBVs received payments for referring or accompanying pregnant women to ANC in the first trimester—or to a control group without incentives. Outcomes such as the proportion of women who initiated ANC in the first trimester, the number of women referred or accompanied by CBVs, and the rate of facility-based deliveries are measured.

The findings reveal that treatment had no significant impact on the proportion of women who initiated ANC in the first trimester. It resulted in a visible increase in CBVs who accompanied women to care in this period by about 29%, though. The intervention also brought with it a notable improvement in the institutional delivery rate, revealing that although financial incentives are incapable of changing early ANC initiation, they are very much able to facilitate supportive behaviors and institutional births.

2. Dependent Variables

The following are the outcome variables the study primarily focuses on.

- **stanccoverage1sttrimester -** Proportion of women seeking ANC in the first trimester
  This variable appears in the regression tables. It is the primary outcome the study aims to influence through the intervention.
- **institutionaldeliverycoverage** - Proportion of women giving birth at health facilities
  This variable appears in the regression tables as a secondary outcome. This variable reflects delivery behavior potentially influenced by increased ANC visits.
- **referred -** Number of women referred to seek ANC in the first trimester by SMAG/NHC
  This variable appears in the NHC-level regression table (Table 5) as a second outcome. It is used as a dependent variable as it is expected to change because of the treatment.
- **accompanied** - Number of women accompanied to seek ANC in the first trimester by SMAG/NHC
  This variable appears in the NHC-level regression table (Table 5) as a second outcome. It is also a direct behavior that the intervention was designed to impact, making it a dependent variable.
- **total** – Total women referred or accompanied to seek ANC by SMAG/NHC
  This variable appears in the NHC-level regression table (Table 5) as a second outcome. Because it summarizes the target behaviors of the intervention, it is naturally considered

the dependent variable.

3. Independent Variables

The following are the factors hypothesized to influence the dependent variables in the study.

- **treatment** – Treatment status (monetary incentive vs. control)
  This variable appears in all regression tables as the main predictor. It represents the randomly assigned intervention and is used to estimate the causal impact of monetary incentives on ANC behaviors. Because it is manipulated by the researchers and is central to the study design, it is the primary independent variable.
- **post** – Dummy variable for the post-baseline period
  This variable is included in Table 4 both as a main effect and in an interaction term with treatment. It is used to test how outcomes change after the treatment period begins and whether the treatment effect varies over time. It is treated as an independent variable on account of its role in predicting and potentially modifying outcomes.
- **femalenhc_ratio** – Female ratio in NHC
  This variable appears in Table 6 both as a main effect and in interaction with the treatment variable. The study takes interest in whether the gender composition of Neighborhood Health Committees influences the outcome, making it an independent variable.
- **femalesmag_ratio** – Female ratio in SMAG
  This variable appears in Table 6 as both a main effect and as part of an interaction term with treatment. It is used to examine whether the proportion of women in Safe Motherhood Action Groups affects outcomes or alters how treatment works. It is treated as an independent variable because it helps explain or modify effects on outcomes.

4. Covariates

The following are the variables that are not the primary focus of the study and included to control for potential confounding and.

- **distance_dmo** – Distance to District Medical Office
  This variable is used in matching and regression adjustment, as noted in Table 3 (“All variables used for pairwise matching are included as controls.”). It’s included to account for structural differences across facilities, which is typical of a covariate.
- **no_maternity_beds** – Number of maternity beds
  This variable is used in matching and regression adjustment, as noted in Table 3 (“All variables used for pairwise matching are included as controls.”). Like the others used in matching, it is included to control for background characteristics that might otherwise bias estimates of treatment effects — making it a covariate.
- **pop_catchment** – Population in the facility’s catchment area
  This variable is used in matching and regression adjustment, as noted in Table 3 (“All

variables used for pairwise matching are included as controls."). It is treated as a covariate to adjust for structural characteristics that could affect outcomes.

- **pop_total** – Population in the NHC's catchment zone
  This variable appears as a control in NHC-level models in Table 5, where notes say "Controls included are training group (dummy variable with 3 categories), population in facility catchment area". It is included as a covariate to reduce potential bias in estimating treatment effects.
- **training** – Orientation team category
  This variable is described in Table 5 notes as a "dummy variable with 3 categories" included as a control.
  It is treated as a covariate to control for variation in how the program may have been introduced, used to improve estimation accuracy rather than as a predictor of interest.
- **pair** – Facility pair identifier
  This variable is referenced in the table notes as part of the matched design: "Results are adjusted for matched pairs, matched dummies and controls not reported" (Table 3 notes).
  It is incorporated as a covariate to control for inherent similarities within each pair.

5. Matching Criteria Evaluation

The matching criteria in this study were based on three structural characteristics of health facilities: (1) **distance_dmo -** distance to the District Medical Office (DMO), (2) **no_maternity_beds -** number of maternity beds, and (3) **pop_catchment -** population in the facility's catchment area. Matched pairs were created using these variables.

From our perspective, it is an overall good choice for the following reasons. To start with, it focused on supply-side factors that are likely to shape maternal healthcare use. Distance from the DMO may influence supervision and assistance, several maternity beds reflect hospital capacity, and the size of the catchment population predicts potential demand. Matching on these characteristics before randomization helped reduce confounding, increase the precision of comparisons between treatment and control groups, and increase statistical power within a relatively small sample of 98 facilities. Furthermore, a minimum distance of 8 km between facilities in a pair was imposed to reduce contamination risk. This mandatory geographic separation step further protects the internal validity of the treatment effect. Additionally, the optimal greedy algorithm was used for pairwise matching in this study, which ensured a systematic and statistically rigorous matching process.

However, some limitations should be acknowledged. By focusing only on structural variables, the matching process did not account for demand-side factors such as cultural attitudes or community education levels. Despite this, given the constraints of working in the field, the trade-offs made were reasonable, and the matching design remains a well-executed and valuable aspect of the study.

Overview of Project 2

1. Brief Description

This study investigates whether providing supplementary drinking water to schoolchildren in water-scarce settings impacts their hydration status and cognitive performance. Using a crossover trial design, 107 students in grades 5-6 from two rural primary schools in Mali participated in two testing conditions: a control day (normal school conditions) and a water intervention day (receiving a 1.5-liter water bottle that was refilled as needed). Outcomes measured included afternoon hydration status through urine samples and performance on six cognitive tests assessing visual attention, visual memory, short-term memory, and visuomotor skills.

The findings reveal that surprisingly few students (only 3%) were dehydrated at the beginning of the school day regardless of condition. Dehydration increased throughout the day under both conditions, but students had significantly better hydration (lower urine specific gravity) in the afternoon when provided with water. Regarding cognitive performance, the intervention showed mixed results with a significant positive effect on reverse number recall but a significant negative effect on line tracing. However, these findings were largely overshadowed by a substantial "practice effect" where students performed significantly better on their second testing day, regardless of water provision. This suggests that while supplementary water measurably improves hydration in the short term, its immediate effects on cognitive performance remain unclear due to methodological challenges in cognitive testing in this population.

2. Dependent Variables

The following are the outcome variables the study primarily focuses on:

Cognitive Performance Measures:

- **LCS** - Letter cancellation score
  This variable appears in Tables 2 and 3 as a measure of visual attention. It represents students' ability to identify target letters among distractors and was hypothesized to improve with better hydration based on prior research.
- **IDS** - Direct image difference score
  This variable appears in Tables 2 and 3 as another measure of visual attention. It tests students' ability to identify differences between simultaneously presented images and was expected to be enhanced by improved hydration.
- **IIS** - Indirect image difference score
  This variable appears in Tables 2 and 3 as a measure of visual memory. It assesses students' ability to remember an image and identify differences when shown a second image, representing a cognitive function potentially sensitive to hydration status.
- **NFC/NFM** - Forward number recall (total correct/maximum span)
  These variables appear in Tables 2 and 3 as measures of short-term memory. They test students' ability to remember and reproduce sequences of numbers and were included based on prior research showing hydration effects on memory performance.

- **NRC/NRM** - Reverse number recall (total correct/maximum span)
  These variables appear in Tables 2 and 3 as measures of working memory and executive function. They represent a more complex cognitive task requiring mental manipulation of information and showed one of the few significant positive associations with water provision.
- **LT** - Line tracing score
  This variable appears in Tables 2 and 3 as a measure of visuomotor skills. It tests students' fine motor control and coordination and shows a significant but unexpectedly negative association with water provision.

3. Independent Variables

Based on the study's analysis approach and the tables you've shared, the independent variables used in the study were:

3.1 Treatment Condition:

- **WATER -** Condition indicator (0=Control 1=Water)
  This variable appears in Table 3 as the main predictor in the first set of regression models. It represents the experimentally manipulated condition where students either received a 1.5L refillable water bottle (1) or experienced normal school conditions (0). Because it is the primary manipulation by the researchers and central to the study design, it is the main independent variable used to test whether water provision affects cognitive performance and hydration status.

3.2 Hydration Status:

- **dehydPM -** Dehydrated in afternoon (0=no 1=yes)
  This variable appears in Table 3 in the middle columns as a predictor of cognitive outcomes. It indicates whether a student was classified as dehydrated (specific gravity ≥ 1.019) in the afternoon. It is treated as an independent variable to test whether the physiological state of dehydration, regardless of how it was achieved, directly impacts cognitive performance. This allows researchers to disentangle whether cognitive effects are due to the act of drinking water or specifically to hydration status.
- **sgpm -** Specific gravity in AFTERNOON
  This variable appears in Table 3 in the rightmost columns as a predictor of cognitive test scores. It represents the actual specific gravity measurements. It is used as an independent variable to test whether there is a dose-response relationship between the degree of hydration/dehydration and cognitive performance. Researchers can, therefore, figure out how much hydration is needed to improve cognition.

4. Covariates

The following are the variables that are not the primary focus of the study and included to control for potential confounding.

- **visit -** Visit day (1=1st visit 2=2nd visit)

This variable is included in all regression models of Table 3. It was included to control for the substantial "practice effect" observed in cognitive testing, where students performed significantly better on their second testing day regardless of treatment condition. Therefore, it is a very crucial covariate for the whole experiment.

- **Grade** - Student grade level (5th or 6th)
  This variable appears specifically in the models for direct and indirect image difference tests in Table 3. The researchers included grade as a control variable in these specific models after identifying it as a significant confounder ($p<0.001$ and $p=0.002$, respectively). It functions as a covariate because developmental differences between 5th and 6th graders could independently affect both cognitive performance and responses to hydration interventions.
- **sex** - Student gender
  This variable was tested as a potential effect modifier, as described in the methods section, and was treated as a covariate for certain analyses. Although not shown in Table 3, sex was found to significantly modify the relationship between afternoon dehydration and forward number recall maximum digit span test scores, leading to stratified analyses. It was included because physiological responses to hydration may differ between males and females, making it a typical covariate.
- **Variables tested but not included in final models**: According to the authors, they mentioned that they first tested potential confounders, including reported drinking in the morning, reported eating in the morning, reported thirst, and morning hydration. But they are not included in the final models.
  - **P107** - Drank anything this morning (0=No 1=Yes)
  - **P106** - Ate food this morning (0=No 1=Yes)
  - **Thirsty** - Self-reported thirst
  - **sgam** - Morning hydration status
  - **dehydAM** - Dehydrated in morning (0=no 1=yes)

5. Crossover design evaluation

From our perspective, the choice of a crossover design for this experiment is a good one overall. First of all, the authors explicitly state that they selected this design over other randomized controlled trial designs due to logistical challenges in randomizing water distribution within classrooms. In a crowded school environment with limited time, it would be impractical to prevent students in the treatment and control groups from sharing water. Other than limited time, conducting the experiment in rural Mali presents resource constraints. Collecting data from 107 students, each serving as their own control, can greatly improve statistical power. Based on this, because each participant serves as their own control, this can effectively control for individual differences, especially for cognitive tests, controlling for individual baseline performance is valuable. Besides, the study also addresses carryover effects by adding a three-day separation between visits, counterbalancing the order of conditions between schools.

Another critical advantage is ethical consideration. Such a design ensures that each student receives supplementary water for one day, whereas a randomized controlled trial would require a control group that does not receive water at all.

However, some limitations should be acknowledged. The most salient problem is that the practice effect's impact on performance of cognitive testing. Students performed dramatically better on their second testing day regardless of water condition. Such practice effects can cover the treatment effect, making it difficult to isolate the true impact of water provision on cognition. But overall, the crossover design is a pragmatic and statistically sound approach that balances scientific rigor with limited resources.

Experimental Design

For this report, we chose Project-2 for our future experimental analysis. Based on the structure and variables of the study, we proposed the following two hypotheses to explore the effects of hydration interventions on cognition and designed corresponding experimental models to test each hypothesis.

Hypothesis 1: Crossover Design

1. Hypothesis

Water intervention improves short-term memory performance as measured by Number Recall Reverse scores (NRC), when controlling for visit order effects.

2. Difference from Original Study

The original study reported a general trend suggesting that drinking water may improve cognitive performance without specifying which cognitive domains would be most affected. This hypothesis aligns with the original Mali study's focus on water intervention's effect on cognition but specifically targets short-term memory as measured by the Number Recall Reverse (NRC) test scores rather than examining multiple cognitive domains simultaneously. By focusing on a single cognitive domain with a clearly defined measure, we can more precisely test the relationship between hydration and this specific aspect of cognitive performance, potentially revealing effects that might be diluted when analyzing multiple cognitive domains together.

3. Design Chosen: Crossover Design

We utilize the crossover design from the original Mali study, where each pupil was tested twice - once under water conditions and once under control conditions, with the order counterbalanced between schools. This design was chosen because it allows each participant to serve as their own control, which eliminates between-subject variability as a source of error and substantially increases statistical power even with a modest sample size. The crossover method also fits perfectly with the methodology of the original study, allowing us to draw directly on

their findings while focusing on the specific cognitive domain of reverse digit recall tests.

4. Variables and Design Setup

- **Independent Variables:**
    - Water intervention (water: 0=control, 1=water condition) - This is our primary treatment, representing whether students received supplementary water or not.
    - Visit days (1=first visit, 2=second visit) - Different from the original model, we've chosen to include visit as a categorical factor and consider its effect rather than controlling it. This allows us to examine the existence and nature of the practice effect directly, rather than just adjusting for it. Modeling the visit as a factor acknowledged that the first and second tests may have qualitatively different effects on performance, especially given the strong practice effect noted in the original study. In addition, it allows us to test the interaction between the water intervention and the order of visits to reveal whether the effects of hydration differed depending on the time of implementation, offering deeper insights into the relationship between hydration and short-term memory performance.
- **Dependent Variable:** NCS **-** Number Recall Reverse correct (raw score)
- **Design Setup:** The study employed a two-period crossover design in which students were tested on two different days. On one day, they received a 1.5-liter bottle of water in the morning and were encouraged to drink throughout the day, with refills provided upon request; on the other, they followed a normal school routine without intervention. The order of treatment and control conditions was counterbalanced between two schools (School A and School B). Cognitive testing was administered at 3:00 p.m. on each visit, focusing on the Reverse Digit Recall Test that measures a student's short-term memory ability. The test is administered under standardized conditions to ensure consistency of measurements between visits.

5. Data Transformation Approach

We will use all the data from both visit days, subsetting the relevant columns including water, visit, and NRC. After initial data exploration, we confirmed there are no missing values in these variables. Further data processing will be needed when introducing a blocking factor and covariate.

The complete case analysis preserves the crossover design structure where each student serves as their own control across the two conditions. We made the following assumptions to ensure valid statistical inference and appropriate interpretation of our results:

- Any carryover effects between visits were minimized by the separation between testing sessions.
- The effect of water intervention can be isolated from the effect of visit order. Our analysis model explicitly includes both factors, enabling us to distinguish between hydration effects and practice effects on cognitive performance.

6. Blocking Only Analysis

Blocking Factor: School (A or B)

We include school as our blocking factor for the crossover design analysis because it represents a fundamental structural element of the original study design. The counterbalancing of treatment was implemented at the school level, with School A receiving water on the first visit and School B receiving water on the second visit. By blocking on school, we can account for potential systematic differences between the two educational environments that might influence cognitive performance, such as teaching resources or student population characteristics. This blocking approach increases the precision of our treatment effect estimate by removing between-school variability from the error term in our analysis.

Further Data Transformation

Besides factors and outcomes (water, visit, NRC), we will now include school in our dataset to account for potential systematic differences between the two schools. Data exploration reveals no missing values in the school variable, allowing us to maintain a complete dataset for our analysis. School is already coded as a categorical variable in the dataset, making it directly suitable for acting as a blocking factor in our ANOVA model.

ANOVA Result

Expectations/Goals: The goal of the analysis is to test whether the water intervention significantly improves short-term memory performance, as measured by NRC scores, while accounting for practice effects in the crossover design. The C(visit) term adjusts for practice effects embedded in the design, and the interaction term adjusts for whether the impact of water differs depending on when it is administered, helping clarify the treatment effect in the presence of order-related variation. School is included as a blocking factor to control for structural differences in the learning environment so that the treatment effects can be more accurately estimated.

Model Used: NRC ~ C(water) * C(visit) + C(school)

Outcomes/Interpretation: According to Table 1, we found the following specific analysis results based on the two-way ANOVA with school as a blocking factor. These results help isolate the treatment effect of water intervention on short-term memory performance while accounting for visit order and school-level differences. The blocking factor is used for better randomization, needs to be designed into the model independent of the actual results, and doesn't need to be significant for blocking to have worked, thus we do not address it in the following interpretations.

- **Water intervention effect C(water), p = 0.007:** The significant p-value indicates that students who received water performed significantly better on the reverse number recall (NRC) task compared to when they didn't receive water. This supports the hypothesis that hydration positively impacts short-term memory.
- **Visit order effect C(visit), p = 0.732:** The visit variable was intended to capture any

"practice effect," where students might score better on the second attempt simply because they were more familiar with the tasks. However, the non-significant p-value indicates that visit order did not have a meaningful influence on NRC scores in this subset. This result is somewhat surprising, as the original study did report substantial practice effects overall. One possible explanation is that for the specific cognitive domain of short-term memory (as measured by NRC), practice effects may be less pronounced than for other cognitive tasks included in the original study. Importantly, by including visit in our model, we were able to properly isolate and test for this potential confounder, allowing us to more confidently attribute performance differences to the water intervention rather than testing sequence.

- **Interaction effect between water and visit C(water):C(visit), p = 0.427:** The interaction tested whether the effect of water varied depending on whether it was administered during the first or second visit. The non-significant result indicates that water had a consistent effect across both visits. In other words, the benefit of water on NRC scores did not depend on test order. This strengthens the case for a true treatment effect rather than an order-dependent one and supports the assumption that the 3-day washout period in the original study was effective in preventing carryover effects.

Table 1: *ANOVA Results for Hypothesis 1 (Blocking Only)*

	df	Sum of Squares	F-value	p-value
C(water)	1.0	21.141	7.469	0.007
C(visit)	1.0	0.333	0.118	0.732
C(school)	1.0	17.069	6.031	0.015
C(water):C(visit)	1.0	1.791	0.633	0.427
Residual	210.0	594.375	NaN	NaN

Post-hoc Test Results

To further explore how the water intervention and visit order influenced NRC scores, we conducted a Tukey HSD post-hoc test (Table 2). This allowed for multiple pairwise comparisons between all combinations of water and visit conditions while adjusting for the family-wise error rate at $\alpha = 0.05$. These comparisons help identify exactly which groups differ from each other beyond what the overall ANOVA reveals.

Key findings from the post-hoc test:++

- **Only one pairwise comparison was statistically significant:** According to Table 2, the comparison between 0_1 (control condition on the first visit) and 1_1 (water condition on the first visit) showed a mean difference of 1.066, with a p-value of 0.007, which is below 0.05. This result confirms that students who received water on their first visit performed significantly better on NRC tasks than those who did not. It highlights a clear benefit of hydration when students have no prior exposure to the cognitive task.
- **No other group differences reached significance:** While some pairwise differences were in the expected direction, their p-values exceeded 0.05. This means that for comparisons like water on the second visit vs. control on the second visit, or water across different visits, the performance differences weren't statistically reliable based on the current data.

The post-hoc analysis strengthens the overall conclusion from the ANOVA. It confirms that water provision has a statistically significant positive effect on NRC scores but most clearly when students receive water during their first exposure to the task. While no other comparisons were statistically significant, the findings suggest that water has a positive effect on cognitive performance.

Table 2: *Post-hoc Tukey Test Results for Hypothesis 1 (Blocking Only): Multiple Comparisons of Means – Tukey HSD (FWER = 0.05)*

group1	group2	meandiff	adjusted p-value	lower	upper	reject
0_1	0_2	0.632	0.219	-0.216	1.481	False
0_1	1_1	1.066	0.007	0.217	1.914	True
0_1	1_2	0.787	0.109	-0.112	1.686	False
0_2	1_1	0.433	0.494	-0.362	1.229	False
0_2	1_2	0.155	0.965	-0.694	1.003	False
1_1	1_2	-0.278	0.831	-1.127	0.570	False

7. Covariate Only Analysis

Continuous Covariate: SGAM (Specific gravity in the morning)

We include morning specific gravity as our covariate not because we're primarily interested in how baseline hydration affects cognitive performance, but because controlling for this variable allows us to better isolate and understand the true effect of water intervention on

short-term memory. This approach is particularly valuable because students' morning hydration levels might vary between testing days for reasons unrelated to our intervention, potentially obscuring the treatment effect we're trying to measure. By statistically adjusting for baseline hydration, we can better estimate the specific impact of the water provision on short-term memory performance.

Further Data Transformation

Besides factors and outcomes (water, visit, NRC), we will now include morning specific gravity (SGAM) in our dataset to account for baseline hydration status. Data exploration reveals no missing values in the SGAM variable, allowing us to maintain a complete dataset for our analysis. SGAM is measured on a continuous scale, with higher values indicating greater dehydration, and will be incorporated directly as a covariate in our ANCOVA model without requiring additional transformation.

ANOVA Results

Expectations/Goals: The goal of the analysis is to test whether the water intervention significantly improves short-term memory performance, as measured by NRC scores, while accounting for variation in students' baseline hydration levels. The variable SGAM (Specific Gravity in the Morning) serves as a continuous covariate that reflects students' hydration status before the intervention. Including SGAM in the model allows us to adjust for pre-existing differences in hydration that could influence cognitive performance. The C(visit) is again included to control for possible practice effects from repeated testing, and the interaction term between water and visit captures whether the effectiveness of the water intervention depends on the order in which it is administered. This covariate-based model improves the precision of treatment effect estimation by reducing unexplained variability tied to physiological differences.

Model Used: NRC ~ C(water) * C(visit) + SGAM

Outcomes/Interpretation: According to Table 3, we found the following specific analysis results based on the two-way ANOVA with SGAM as a covariate.

- **Water intervention effect C(water), p = 0.010:** The water condition remained statistically significant even after adjusting for students' baseline hydration levels. This result improves the previous analysis and indicates that the water intervention had a true effect on improving NRC scores, independent of how hydrated the students were at the start of the day.
- **Visit order effect C(visit), p = 0.451:** As with the blocking-only model, visit order was not significant. This again suggests that practice or test familiarity did not meaningfully affect performance on the NRC task.
- **Interaction effect C(water):C(visit), p = 0.062:** This interaction was not statistically significant at the 0.05 level, though it came relatively close. This could indicate a weak trend where the effect of water may differ slightly based on visit sequence, but further data would be needed to confirm this.

- **SGAM covariate, p = 0.756:** SGAM was not statistically significant in the model. This suggests that students' hydration status in the morning did not have a strong independent effect on their NRC performance, nor did it substantially moderate the water intervention's impact. However, its inclusion still helped control for baseline variability and added rigor to the analysis.

Table 3: *ANOVA Results for Hypothesis 1 (Covariate Only)*

	df	Sum of Squares	F-value	p-value
C(water)	1.0	19.049	6.701	0.010
C(visit)	1.0	1.620	0.570	0.451
C(water):C(visit)	1.0	9.994	3.516	0.062
SGAM	1.0	0.275	0.097	0.756
Residual	209.0	594.100	NaN	NaN

Post-hoc Test Results

To further examine how water intervention and visit order jointly influenced NRC scores, we conducted a Tukey HSD post-hoc test. This test compared all pairwise combinations of the water and visit groups while adjusting for the family-wise error rate at $\alpha = 0.05$. These comparisons provide a more granular view of where significant differences in short-term memory performance occur, beyond what the overall ANOVA reveals.

According to Table 4, we found the following specific analysis results from the post-hoc comparisons:

- **Only one pairwise comparison was statistically significant:** The only statistically significant difference occurred between the groups 0_1 (control condition on visit 1) and 1_1 (water condition on visit 1), with a p-value of 0.007. This improves our earlier finding that students who received water during their first test session performed significantly better on the NRC task than those who did not.
- **All other pairwise comparisons were not statistically significant**: Although some comparisons showed trends in the expected direction, they did not meet the threshold for statistical significance.

The post-hoc analysis confirms the central finding that water significantly improved NRC performance when administered during the first session. This effect persists even after adjusting for baseline hydration status using SGAM, indicating that the treatment effect is not simply an artifact of pre-existing differences in hydration. Other pairwise comparisons did not reach significance, likely due to variability or sample size limitations, but the primary result is both statistically and educationally meaningful.

Table 4: *Post-hoc Tukey Test Results for Hypothesis 1 (Covariate Only): Multiple Comparisons of Means – Tukey HSD (FWER = 0.05)*

group1	group2	meandiff	adjusted p-value	lower	upper	reject
0_1	0_2	0.632	0.219	-0.216	1.481	False
0_1	1_1	1.066	0.007	0.217	1.914	True
0_1	1_2	0.787	0.109	-0.112	1.686	False
0_2	1_1	0.433	0.494	-0.362	1.229	False
0_2	1_2	0.155	0.965	-0.694	1.003	False
1_1	1_2	-0.278	0.831	-1.127	0.570	False

Hypothesis 2: Factorial Design

1. Hypothesis

Water intervention and morning eating status jointly influence visual attention performance, as measured by Letter Cancellation scores (LCS). We hypothesize that both drinking water and eating breakfast positively affect cognitive performance by solving hydration and providing necessary energy, respectively.

We further expect an interaction effect: The impact of providing water on students' LCS scores may vary depending on whether students ate breakfast or not, as those who ate may be better primed to benefit cognitively from hydration, potentially amplifying the effect of the water intervention compared to students who skipped breakfast.

2. Difference from Original Study

This hypothesis differs substantially from the original studies by examining the interaction between experimental water provision and pre-existing consuming behaviors (morning eating). The original studies focused solely on the direct effect of water provision without considering how its effects might be modified by students' prior consumption practices. Studying these 2 factors together is particularly meaningful because nutrition and hydration

represent complementary physiological needs that together support cognitive function. Additionally, we specifically focus on visual attention performance rather than general cognitive effects.

3. Design Chosen: Factorial Design

We implement a factorial design to examine how two factors (water intervention and morning eating) interact to affect cognitive performance. This design was chosen because it allows us to examine not only main effects but also interaction effects between 2 factors, potentially uncovering mechanisms that may not be found when the factors are studied individually. This factorial design acknowledges that water supply may affect students differently depending on their pre-existing eating behavior. It provides a robust framework for understanding heterogeneous treatment effects, allowing us to identify the specific conditions under which water interventions may be most beneficial for cognitive performance and providing an empirical basis for subsequent, more precise interventions to help students.

4. Variables and Design Setup

- **Independent Variables:**
    - Water intervention (water: 0=control, 1=water condition) - This is our experimental manipulation
    - Morning eating status (P106: 0=did not eat food, 1=ate food in the morning) - This self-reported variable captures students' behaviors, which might influence their conditions after the water intervention.
- **Dependent Variable:** LCS - Letter Cancel Score
- **Design Setup:** The study utilizes the same two-visit structure as described in Hypothesis 1. On the water intervention day, each student received a 1.5-liter bottle of water in the morning and was encouraged to drink throughout the day, with refills provided upon request. On the control day, students followed their usual school routine without any additional water provision. The order of conditions was counterbalanced between School A and School B to minimize systematic bias.
  However, the analytical focus shifted to examining the interaction between water intervention and morning eating habits now. Prior to testing on each visit day, pupils reported whether they had consumed anything that morning (P106). The Letter Cancellation test is administered to all pupils during the afternoon cognitive testing session at 3:00 PM. The factorial structure creates four analytical groups: (1) No Breakfast + Control condition, (2) Breakfast + Water intervention, (3) No Breakfast + Control condition, and (4) Breakfast + Water intervention. This allows for the comparison of both main effects and the interaction effect on visual attention performance.

5. Data Transformation Approach

We will use all the data from both visit days to increase statistical power, subsetting the

relevant columns including water, P106, and LCS. After initial data exploration, we confirmed there are no missing values in these variables. Further data processing will be needed when introducing a blocking factor and covariate.

While we are using data from both visits, we are not explicitly modeling the repeated-measures aspect of the crossover design. Instead, we focus on the factorial structure of water intervention and morning eating habits. We made the following assumptions to ensure valid statistical inference and appropriate interpretation of our results:

- Though data is collected from both days, we focus on the influence and interaction of 2 factors. The effects between water intervention and morning eating is of primary interest.
- The factorial effects are not substantially biased by the non-independence of observations. While we acknowledge that observations may come from the same student, the balanced nature of the intervention provision means that each student contributes equally to all factorial combinations across the two visits. This balance helps minimize the impact of non-independence on our primary research question.
- The imbalance between groups reflects a genuine pattern in the population - the majority of students reported eating something in the morning. Though this imbalance partially influences statistical power for detecting interaction effects, We believe that preserving the real behavioral patterns observed in the study population provides more meaningful insights than artificially balancing the groups through making up data or other methods.

6. Blocking Only Analysis

Blocking Factor: Morning dehydration status (dehydAM)

We include morning dehydration status as our blocking factor because it captures a condition before treatment that may introduce substantial variability in cognitive outcomes. By conducting randomization based on whether students were dehydrated or not in the morning, we reduce within-group variability and ensure a more balanced comparison across hydration conditions. Blocking in this way supports more accurate estimation of treatment effects and enhances the reliability of our conclusions.

Further Data Transformation

Besides factors and outcomes (water, P106, LCS), we will now include morning dehydration status (dehydAM) in our dataset to account for pre-existing physiological differences. Data exploration reveals no missing values in the dehydAM variable, allowing us to maintain a complete dataset for our analysis. DehydAM is already coded as a binary indicator (0=no, 1=yes) in the dataset, making it directly suitable for functioning as a blocking factor in our ANOVA model.

ANOVA Results

Expectations/Goals: This analysis will assess whether the factor water, P106, as well as the interaction between water and P106, have a significant impact on the outcome Letter

Cancellation Score (LCS) while using dehydAM as the blocking factor to improve randomization. By controlling for the morning dehydration status, the goal is to reduce random variation and isolate the effects of the main treatments more clearly.

Model Used: LCS ~ C(water) * C(P106) + C(dehydAM)

Outcomes/Interpretation: As shown in Table 5, no factor has a significant effect on LCS. Specifically;

- **Water Intervention Effect C(Water), p = 0.667**: the water variable remains non-significant, with an F-value of 0.185 and a p-value of 0.667. This confirms that differences in water treatment do not significantly contribute to the LCS outcome. It shows that student's abilities to identify target letters among distractors won't significantly be improved with better hydration.
- **Program effect C(106), p = 0.738:** Similarly, the P106 variable is non-significant, with an F-value of 0.112 and a p-value of 0.738, indicating that the fact that whether students eat anything in the morning alone does not affect their performance.
- **Interactive effect of water and programs C(water): C(P106), p = 0.349**: Furthermore, the interaction between water and P106 shows an F-value of 0.881 and a p-value of 0.349, indicating their combined effect is not statistically significant. Since these findings persist even after accounting for potential variability introduced by dehydAM, it is concluded that neither the individual treatments (water and eating in the morning) nor their interaction significantly influence the outcome.

The blocking factor is used for better randomization, needs to be designed into the model independent of the actual results, and doesn't need to be significant for blocking to have worked. Thus, we do not address the blocking factor in the above interpretations.

Table 5: *ANOVA Results for Hypothesis 2 (Blocking Only)*

	df	Sum of Squares	F-value	p-value
C(water)	1.0	12.609	0.185	0.667
C(P106)	1.0	7.631	0.112	0.738
C(dehydAM)	1.0	2.528	0.037	0.847
C(water):C(P106)	1.0	59.899	0.881	0.349
Residual	209.0	14209.106	NaN	NaN

Post-hoc Test Results

Although we didn't find a significant effect in ANOVA, we conducted a Tukey HSD post-hoc test to further verify our findings. This test compared all pairwise combinations of the variables water and P106 while adjusting for the family-wise error rate at $\alpha = 0.05$. These comparisons provide a more granular view of each pair's differences beyond what the overall ANOVA reveals. As shown in Table 6 below, there is no significant difference among all groups, echoing our ANOVA result that none of the water, P106, or their interaction has a significant effect on LCS.

Table 6: *Post-hoc Tukey Test Results for Hypothesis 2 (Blocking Only): Multiple Comparisons of Means – Tukey HSD (FWER = 0.05)*

group1	group2	meandiff	adjusted p-value	lower	upper	reject
0_0	0_1	0.443	0.998	-5.861	6.747	False
0_0	1_0	3.789	0.852	-8.393	15.970	False
0_0	1_1	-0.306	0.999	-6.577	5.964	False
0_1	1_0	3.346	0.856	-7.531	14.222	False
0_1	1_1	-0.749	0.920	-3.788	2.290	False
1_0	1_1	-4.095	0.763	-14.952	6.762	False

7. Covariate Only Analysis

Continuous Covariate: Age

Although age is recorded as an integer (in years) rather than a strictly continuous variable, it is commonly treated as continuous in statistical models because cognitive abilities develop gradually across childhood and adolescence.

We include age as a covariate not because we are primarily interested in developmental effects on visual attention, but because adjusting for age allows us to more accurately detect the effects of water intervention and morning eating behaviors. By accounting for age-related variance in Letter Cancellation Scores (LCS), we reduce the background noise through better randomization. This is especially important given the wide age range in our sample (9–16 years), which could otherwise obscure more subtle effects of hydration-related factors. Including age in our model helps more accurately estimate how 2 factors influence the outcome, which is the central focus of our analysis.

Further Data Transformation

Besides factors and outcomes (water, P106, LCS), we will now include age in our dataset

to account for developmental differences in visual attention. Data exploration reveals 56 missing values in the age variable, requiring data cleaning. After careful consideration, we decided to exclude these cases from our analysis. The reasons are as follows.

To start with, the original study mentions the age distribution of students "A total of 120 pupils in grades five (ages 9–13) and six (ages 10–16) were recruited," indicating a wide age range even within the same grade. This variability suggests that imputing age using the overall mean value or grade-level means would likely introduce substantial bias and fail to reflect the true developmental variation among students.

Furthermore, we conducted a power analysis to check if adequate statistical power would remain after excluding these cases. Using established statistical methods, we selected a medium effect size (Cohen's d = 0.5), which reflects meaningful but not overwhelmingly large improvements in cognitive performance due to water intervention. This effect size aligns with expected outcomes in educational intervention research and accounts for typical variability in student responses. We set the significance level (α) at the conventional threshold of 0.05. This level maintains a balance between statistical rigor and practical decision-making in real-world school settings. To minimize Type II errors, we specified a power level ($1 - \beta$) of 0.8, ensuring an 80% probability of detecting a true effect if it exists.

After inputting the key factors—effect size (0.5), significance level (alpha = 0.05), and desired power (0.8)—we conducted a power test for our 2-way factorial ANOVA using the solve_power function. The analysis determined that a total sample size of approximately 48 participants is required to detect medium effect sizes with 80% power at $\alpha = 0.05$. The remaining sample would be 158 observations if cases with missing age are excluded, which exceeds the requirement, indicating still sufficient statistical power for our analysis.

To sum up, we used the original data of 2 visits from the study and performed data cleaning by excluding cases with missing age values. This approach is reasonable because of the wide variation in age across participants and the confirmation from our power analysis that the remaining sample size is sufficient to support valid statistical conclusions.

ANOVA Results

Expectations/Goals: The covariate analysis will assess whether the factor water (P106) and their interaction significantly affect performance as measured by the LCS, while controlling for the age of participants as a covariate. By adjusting for age, the variance due to individual differences could be effectively reduced and hence better isolate the effects of the major experimental treatments.

Model Used: LCS ~ C(water) * C(P106) + Age

Outcomes/Interpretation: Findings were observed and guided our conclusions from the two-way ANOVA analysis with age as a covariate (shown in Table 7). The results isolate the treatment effect of water and program P106 (eating breakfast) on performances while adjusting for age differences. The analysis has shown that hydration has a significant impact, however, the

behavior of having breakfast solely, the interaction of water and breakfast, nor age solely do not explain additional variance in scores.

- **Water intervention effect C(water), p = 0.026:** The significant p-value of the water effect shows that students who had water performed significantly better on LCS tasks than students who did not. This finding supports the hypothesis of the positive influence of hydration on attention or visual scanning abilities.
- **Program effect C(P106), p = 0.316:** The insignificant p-value of the program effect shows that students' LCS performance has no impact regardless of eating in the morning. This concludes that the educational intervention alone does not improve cognitive tasks.
- **Interactive effect of water and program C(water):C(P106), p = 0.162:** The insignificant interaction of water and breakfast has shown that the combination does not create additional enhancements beyond each individual impact. The effect of hydration remains consistent regardless of whether you eat breakfast in the morning or not.
- **Covariate effect Age, p = 0.742:** With an insignificant p-value, age is shown not to have a sole effect on LCS scores. However, age as a covariate does help adjust for the specific age-related variability, which improves the precision of the main treatment estimates, especially for the water effect.

Table 7: *ANOVA Results for Hypothesis 2 (Covariate Only)*

	df	Sum of Squares	F-value	p-value
C(water)	1.0	277.581	5.059	0.026
C(P106)	1.0	55.618	1.014	0.316
C(water):C(P106)	1.0	108.582	1.979	0.162
Age	1.0	5.992	0.109	0.742
Residual	153.0	8395.136	NaN	NaN

Post-hoc Test Results

Due to the finding of the main effect of water intervention, we conducted the post-hoc test to further explore potential pairwise differences.

The post-hoc analysis compared all pairwise combinations of the variables water and P106 while selecting age as a covariate and adjusting for the family-wise error rate at $\alpha = 0.05$. Since all adjusted p-values exceed 0.05 shown in Table 8, we can conclude that no specific group pairing demonstrated significant differences. However, despite no pairwise comparison alone driving this significance, the overall water treatment showed a main effect. This supports the broader conclusion that “providing water improves letter cancellation scores,” even though

group-specific comparisons were not significant after adjusting for multiple comparisons.

Table 8: *Post-hoc Tukey Test Results for Hypothesis 2 (Covariate): Multiple Comparisons of Means – Tukey HSD (FWER = 0.05)*

group1	group2	meandiff	adjusted p-value	lower	upper	reject
0_0	0_1	12.897	0.309	-6.409	32.204	False
0_0	1_0	10.0	0.687	-13.495	33.495	False
0_0	1_1	10.026	0.534	-9.282	29.334	False
0_1	1_0	-2.897	0.947	-16.635	10.840	False
0_1	1_1	-2.872	0.078	-5.953	0.210	False
1_0	1_1	0.026	1.0	-13.714	13.766	False